The Chinese University of Hong Kong, Shenzhen

Course Outline

1. Course identity

A. Course as listed at CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	DDA4080
Course title (English)	Capstone Project
Course title (Chinese)	毕业设计
Units	3
Description (English)	This course offers students a hands-on experience to apply their data science skills in real-world scenarios. Through this course, students work on practical projects, utilizing various techniques, including computer programming and software engineering, machine learning algorithms, statistical models, etc. to solve complex problems across various domains. Emphasizing interdisciplinary collaboration, students gain teamwork experience while leveraging diverse perspectives to address multifaceted challenges. By the course's end, students will gain experience in industry-standard tools and methodologies.
Description (Chinese)	该课程为学生提供在现实场景中应用数据科学技能的动手实践机会。注册此课程的学生将组队参与实践项目，运用计算机编程与软件工程、机器学习算法、统计模型等方面的知识解决众多领域中的复杂问题。课程强调跨学科合作，学生在团队合作中积累经验，同时利用多样化的视角解决多维度的挑战。在课程结束时，学生将获得行业标准工具和方法论相关经验。

B. Corresponding course at CUHK

Please give details of the *closest* corresponding course at CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course at CUHK, please make multiple copies of the block below.

Course code	
Course title (English)	
Course title (Chinese)	
Units	
Description (English)	

Description (Chinese)	

2. Course requisites

Please state prerequisites and co-requisites, in terms of courses at CUHK-Shenzhen or any other requirements.

A. Prerequisites

Refer to the specific requirements of each project

B. Co-requisites

C. Others

3. Learning outcomes

Upon completing this course, students will be able to:
Implement a project or part of a project by connecting what they have learnt in other courses
Have a high-level knowledge of project implementation in a team work environment.

4. Course syllabus

Please highlight fundamental concepts involved in each topic to help students better understand what is covered in the course.

The student needs to apply, after which the professor supervisor and practitioner supervisor will select and form teams for each project based on its specific requirements. Each team needs to submit a biweekly update (either written or oral), a midterm report (either written or oral), and a final report (written), and deliver a final presentation.

5. Assessment scheme

Component/ method	**% weight**
Progress Performance (biweekly report)	30
Midterm Report	30
Final Report and Presentation	40

6. Grade descriptor

The grade descriptor should align with specific course learning outcomes, not a generic description applicable to all courses.

Type:**General**

Grade	Description
A	The project is successfully completed on schedule with excellent teamwork; all project deliverables were met and even exceeded.
B	The project is mostly completed on schedule with good teamwork; all major project deliverables were met.
C	The project is partially completed with acceptable teamwork; some of the project deliverables were met.
D	The project is not completed, but good efforts, acceptable teamwork.
F	The project is not completed, and no good efforts, acceptable teamwork.

7. Feedback for evaluation

Please specify forms of evaluation to generate feedback on classes from students.

Email to professor supervisor

8. Readings

9. Course components

Activity	Hours/week
Project-based learning in teams with two supervisors	20

10. Indicative teaching plan

Week	Content/ topic/ activity
N/A	N/A

11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections. Example: 150 students for lecture (x 2); 30 students for tutorials (x 10).

2-6 students per team

12. Academic honesty and plagiarism

A course outline may include subject-specific requirements on plagiarism, in addition to the University established policies.

Students must complete all coursework and assessments independently, unless AI tool use is explicitly permitted. Unauthorized or improper use of AI may constitute academic dishonesty and harm learning. Misuse includes submitting AI-generated work as one's own, using AI to cheat, or employing AI unethically. Always obtain permission before using AI in academic work, and be sure to acknowledge any AI use in submissions.

13. Approval

Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences? Have the details (as in this document) been approved at School or other level in CUHK- Shenzhen?

14. Any other information

15. Version date

Version number	2
As of (date)	2025/11/11